Parameter Calculation Derivation

Neurons per hidden layer: 4
Neurons in output layer: 1

1. 2-Layer Network (Architecture: 2 -> 4 -> 1)

Layer 1 (Hidden): (2 inputs * 4 neurons) + 4 biases = 12 parameters
Layer 2 (Output): (4 inputs * 1 neuron) + 1 bias = 5 parameters
Total Parameters: 12 + 5 = 17

2. 5-Layer Network (Architecture: 2 -> 4 -> 4 -> 4 -> 4 -> 1)

Layer 1 (Hidden): (2 inputs * 4 neurons) + 4 biases = 12 parameters
Layers 2, 3, and 4 (Hidden): 3 layers * ((4 inputs * 4 neurons) + 4 biases) = 3 * 20 = 60 parameters
Layer 5 (Output): (4 inputs * 1 neuron) + 1 bias = 5 parameters
Total Parameters: 12 + 60 + 5 = 77

3. 10-Layer Network (Architecture: 2 -> 4 (x8) -> 1)

Layer 1 (Hidden): (2 inputs * 4 neurons) + 4 biases = 12 parameters
Layers 2 through 9 (Hidden): 8 layers * ((4 inputs * 4 neurons) + 4 biases) = 8 * 20 = 160 parameters
Layer 10 (Output): (4 inputs * 1 neuron) + 1 bias = 5 parameters
Total Parameters: 12 + 160 + 5 = 177